

Reproduction of the Two-Phase Non-Charring Ablation Solution

1 Paper Objective and Reproduction Scope

Ablative thermal protection removes energy by sacrificing material at a heated surface. Numerical material-response codes can represent this process, but they are difficult to verify when thermal conductivity, specific heat, density, phase change, and surface recession are coupled. Kato and Matsuda address this problem by deriving exact quasi-steady temperature relations for a semi-infinite, one-dimensional, non-charring ablator with temperature-dependent properties [1]. Their stated purpose is to provide a rapid engineering estimate, clarify the behavior of the reduced model, and supply a reference solution for checking numerical codes.

The paper treats a single-phase solid and a two-phase Teflon model. This reproduction focuses on the two-phase model used in Figures 6 and 7. The model contains an amorphous region next to the heated surface and a crystalline region farther into the material. It includes surface recession, latent heat, and different temperature-dependent properties in the two regions. It does not include char formation or transport of pyrolysis gas.

Figure 6 compares the analytical profile with a finite-difference result for one surface temperature and recession velocity. Figure 7 holds the recession velocity fixed and varies the surface temperature. The reproduction evaluates the analytical integrals, solves the original differential equation by two independent numerical methods, checks physical properties and limiting cases, and overlays the calculated curves on the published figures.

2 Moving-Coordinate Energy Equation

Let y denote a stationary coordinate measured from the initial surface, and let $S(t)$ be the recession distance. The coordinate fixed to the receding surface is

$$y = x + S(t), \quad V = \frac{dS}{dt}. \quad (1)$$

The stationary one-dimensional heat equation is

$$\rho(T)C_p(T) \left. \frac{\partial T}{\partial t} \right|_y = \frac{\partial}{\partial y} \left(k(T) \frac{\partial T}{\partial y} \right). \quad (2)$$

At fixed x , the chain rule gives

$$\left. \frac{\partial T}{\partial t} \right|_x = \left. \frac{\partial T}{\partial t} \right|_y + V \frac{\partial T}{\partial x}. \quad (3)$$

Because spatial derivatives in x and y are equal at a fixed time, substitution of equation (3) into equation (2) gives

$$\rho C_p \left. \frac{\partial T}{\partial t} \right|_x = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \rho C_p V \frac{\partial T}{\partial x}. \quad (4)$$

The quasi-steady assumption sets the time derivative at fixed x to zero, so

$$\frac{d}{dx} \left(k(T) \frac{dT}{dx} \right) + V \rho(T) C_p(T) \frac{dT}{dx} = 0. \quad (5)$$

Define the volumetric sensible-enthalpy difference from a reference temperature T_r as

$$H(T; T_r) = \int_{T_r}^T \rho(u) C_p(u) du. \quad (6)$$

Since $dH/dx = \rho C_p dT/dx$, equation (5) can be written

$$\frac{d}{dx} \left[k(T) \frac{dT}{dx} + V H(T; T_r) \right] = 0. \quad (7)$$

Its first integral is

$$k(T) \frac{dT}{dx} + V H(T; T_r) = C. \quad (8)$$

The script `symbolic_derivation.py` differentiates (8) and verifies exact recovery of (5).

3 Crystalline-Region Solution

The local crystalline coordinate is $z = x - \delta_m$, where δ_m is the amorphous-layer thickness. The crystalline boundary conditions are

$$T(0) = T_m, \quad T(z) \rightarrow T_0, \quad \frac{dT}{dz} \rightarrow 0 \quad \text{as } z \rightarrow \infty. \quad (9)$$

Taking $T_r = T_0$ in equation (8) and applying the far-field conditions makes $C = 0$. Therefore,

$$k_c(T) \frac{dT}{dz} = -V \int_{T_0}^T \rho_c(u) C_{p,c}(u) du. \quad (10)$$

Separating variables and integrating from T to T_m gives

$$z(T) = \frac{1}{V} \int_T^{T_m} \frac{k_c(v)}{\int_{T_0}^v \rho_c(u) C_{p,c}(u) du} dv. \quad (11)$$

The crystalline depth in the global coordinate is

$$x(T) = \delta_m + z(T). \quad (12)$$

The inward conductive heat flux is defined as $q = -k dT/dx$. Equation (10) therefore gives

$$q_c(T) = V \int_{T_0}^T \rho_c(u) C_{p,c}(u) du. \quad (13)$$

4 Amorphous Region and Phase Interface

The amorphous boundary conditions are

$$T(0) = T_s, \quad T(\delta_m) = T_m. \quad (14)$$

Define the crystalline sensible-energy contribution and phase-change contribution as

$$\phi_1 = \int_{T_0}^{T_m} \rho_c(u) C_{p,c}(u) du, \quad \phi_2 = \rho_m h_m, \quad (15)$$

where the interface density used by the paper is

$$\rho_m = \frac{\rho_a(T_m) + \rho_c(T_m)}{2}. \quad (16)$$

Conservation of energy across the moving phase interface requires

$$q_a(T_m) - q_c(T_m) = V \rho_m h_m = V \phi_2. \quad (17)$$

The amorphous heat flux that satisfies this jump is

$$q_a(T) = V \left[\int_{T_m}^T \rho_a(u) C_{p,a}(u) du + \phi_1 + \phi_2 \right]. \quad (18)$$

Using $q_a = -k_a dT/dx$ and separating variables gives

$$x(T) = \frac{1}{V} \int_T^{T_s} \frac{k_a(v)}{\int_{T_m}^v \rho_a(u) C_{p,a}(u) du + \phi_1 + \phi_2} dv. \quad (19)$$

The layer thickness follows directly from the temperature at the interface,

$$\delta_m = x(T_m) = \frac{1}{V} \int_{T_m}^{T_s} \frac{k_a(v)}{\int_{T_m}^v \rho_a(u) C_{p,a}(u) du + \phi_1 + \phi_2} dv. \quad (20)$$

Equations (19), (20), and (12) define the full two-phase temperature profile. The calculation first evaluates depth as a function of temperature, which is monotone, and then numerically inverts that relation to obtain $T(x)$.

5 Polynomial Evaluation and Scaling

The published property fits are low-degree polynomials. If

$$\rho(T) = \sum_i \rho_i T^i, \quad C_p(T) = \sum_j c_j T^j, \quad (21)$$

then

$$\rho(T) C_p(T) = \sum_n a_n T^n, \quad a_n = \sum_{i+j=n} \rho_i c_j. \quad (22)$$

The inner enthalpy integral is evaluated exactly as

$$H(T; T_r) = \sum_n \frac{a_n}{n+1} (T^{n+1} - T_r^{n+1}). \quad (23)$$

The code forms these coefficients by polynomial convolution. Numerical quadrature is only used for the remaining outer depth integral.

Both depth relations have the form $x(T) = F(T)/V$. Therefore,

$$x(T; \lambda V) = \frac{x(T; V)}{\lambda}, \quad T(x; \lambda V) = T(\lambda x; V). \quad (24)$$

This scaling supplies a test that does not depend on a digitized reference curve.

6 Material Properties and Figure 6 Conditions

Figure 6 uses

$$T_s = 950 \text{ K}, \quad T_m = 600 \text{ K}, \quad T_0 = 300 \text{ K}, \quad V = 7.5545 \times 10^{-5} \text{ m/s}, \quad h_m = 58615 \text{ J/kg}. \quad (25)$$

The amorphous properties are

$$k_a(T) = 0.88090 - 0.0013984T + 5.8197 \times 10^{-7}T^2, \quad (26)$$

$$C_{p,a}(T) = 904.35 + 0.65314T, \quad (27)$$

$$\rho_a(T) = 2070 - 0.7T, \quad (28)$$

and the crystalline properties are

$$k_c(T) = 0.050242 + 0.00061420T, \quad (29)$$

$$C_{p,c}(T) = 514.98 + 1.5629T, \quad (30)$$

$$\rho_c(T) = 2119 + 0.792T - 2.105 \times 10^{-3}T^2. \quad (31)$$

SI units are used throughout. Substitution into equation (20) gives

$$\delta_m = 0.637199 \text{ mm}. \quad (32)$$

The paper reports 0.64 mm. Since that value is rounded, the small difference is not treated as a measured model error.

Figure 6 plots the piecewise profile

$$T(x) = \begin{cases} T_a(x), & 0 \leq x \leq \delta_m, \\ T_c(x - \delta_m), & x > \delta_m, \end{cases} \quad (33)$$

where T_a and T_c are the inverses of equations (19) and (11). The change in slope at T_m follows from the different material properties and the latent heat term in equation (17).

7 Independent Numerical Calculations

The first independent calculation applies a conservative finite-difference form of equation (5). At an interior node,

$$\frac{k_{i+1/2}(T_{i+1} - T_i) - k_{i-1/2}(T_i - T_{i-1})}{\Delta x^2} + (\rho C_p)_i V \frac{T_{i+1} - T_{i-1}}{2\Delta x} = 0, \quad (34)$$

where $k_{i+1/2} = (k_i + k_{i+1})/2$. This residual uses the differential equation rather than the analytical temperature profile. The crystalline domain is truncated at the paper's $L_\infty = 0.05 \text{ m}$.

The second independent calculation writes

$$G = k(T) \frac{dT}{dx} \quad (35)$$

and solves the first-order boundary-value system

$$\frac{dT}{dx} = \frac{G}{k(T)}, \quad (36)$$

$$\frac{dG}{dx} = -\rho(T)C_p(T)V \frac{G}{k(T)}. \quad (37)$$

Adaptive collocation is run with a requested tolerance of 10^{-8} . This provides a tighter numerical comparison than the fixed finite-difference grid while remaining separate from the analytical quadrature and inversion.

8 Symbolic and Test-Driven Verification

The SymPy calculation checks three exact identities. It differentiates the first integral to recover equation (5), reduces the phase-interface energy residual to zero, and differentiates equation (23) to recover $\rho(T)C_p(T)$. These are algebraic checks and do not use floating point tolerances.

Development followed short red–green–refactor steps. The first tests required positive material properties and the published boundary conditions. Later tests added the interface heat-flux jump, the reported layer thickness, the constant-property limit, and the scaling in equation (24). The finite-difference and boundary-value calculations were then added as independent references. Figure 7 ordering, plot generation, and paper overlays were tested last.

The tests remain separated by purpose in `tests/`: material properties, analytical behavior, energy balance, limiting behavior, independent numerical solutions, error metrics, symbolic identities, and generated figures.

9 Figure 7 Surface-Temperature Sweep

Figure 7 uses the same properties, T_m , T_0 , and V , but evaluates the solution at

$$T_s \in \{900, 950, 1000\} \text{ K.} \quad (38)$$

For each surface temperature, equation (20) determines a different amorphous-layer thickness:

T_s (K)	δ_m (mm)
900	0.600428
950	0.637199
1000	0.665689

Increasing T_s extends the amorphous integral over a larger temperature interval and increases δ_m . The crystalline relation $T_c(z)$ is unchanged because T_m , T_0 , V , and the crystalline properties are fixed. Only its origin shifts in the global coordinate through $x = \delta_m + z$. Figure 7 therefore checks both the predicted surface-temperature trend and the translation of the crystalline profile.

10 Verification and Error Metrics

In the constant-property, single-phase limit, the solution reduces to

$$T - T_0 = (T_s - T_0) \exp\left(-\frac{V\rho C_p}{k}x\right), \quad (39)$$

and a round-trip test recovers x from the calculated temperature.

For numerical temperatures $T_i^{(n)}$ and analytical temperatures $T_i^{(a)}$ at $N = 501$ uniformly spaced depths, define

$$e_i = T_i^{(n)} - T_i^{(a)}, \quad (40)$$

$$E_{\max} = \max_i |e_i|, \quad (41)$$

$$E_{\text{RMSE}} = \left(\frac{1}{N} \sum_{i=1}^N e_i^2\right)^{1/2}, \quad (42)$$

$$E_{\text{rel}} = 100 \max_i \left| \frac{e_i}{T_i^{(a)}} \right|, \quad (43)$$

$$E_{\text{NRMSE}} = 100 \frac{E_{\text{RMSE}}}{\max_i T_i^{(a)} - \min_i T_i^{(a)}}. \quad (44)$$

The comparisons are calculated from temperature values rather than extracted from image pixels.

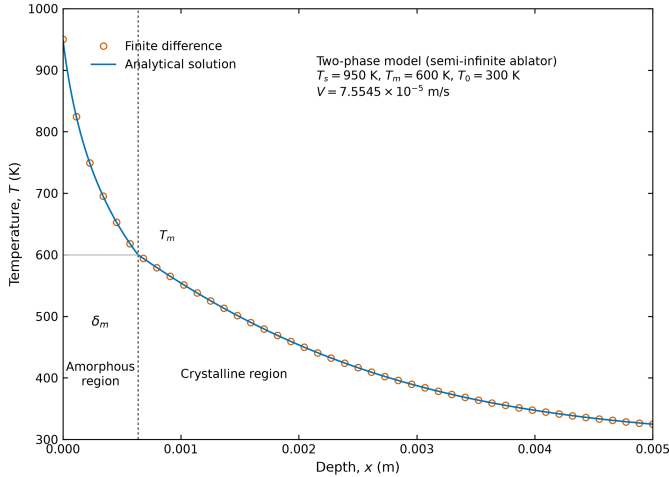
Case	FD max (K)	FD RMSE (K)	FD max (%)	BVP max (K)
Fig. 6, 950 K	0.07535	0.00813	0.00818	8.6×10^{-9}
Fig. 7, 900 K	0.04998	0.00881	0.00558	5.0×10^{-9}
Fig. 7, 950 K	0.09244	0.01223	0.00979	4.9×10^{-9}
Fig. 7, 1000 K	0.15848	0.01898	0.01597	4.8×10^{-9}

The boundary-value calculation is tested with a 2×10^{-8} K maximum-error limit. A 10^{-14} limit would be reasonable for two equivalent algebraic expressions evaluated in double precision, but not for a differential-equation solve requested at a 10^{-8} tolerance.

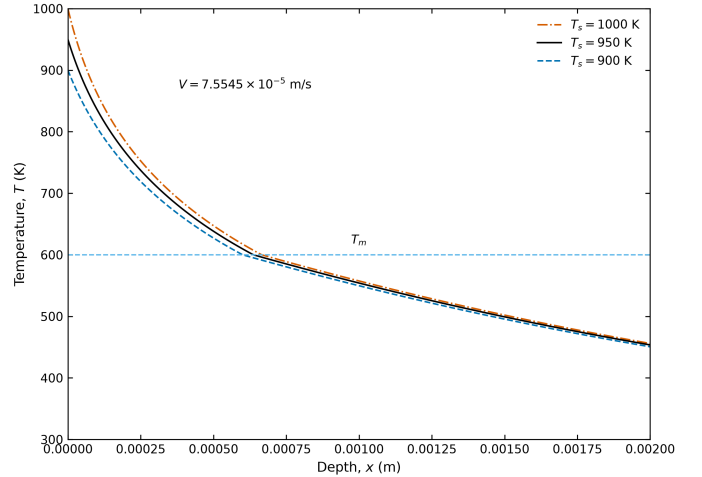
11 Interpretation and Limitations

Figure 6 shows a monotone temperature decrease through both phases. The slope changes at T_m because the properties change and the interface consumes latent heat. The analytical, finite-difference, and boundary-value profiles agree across the plotted domain. Figure 7 shows that increasing T_s raises the profile and increases δ_m ; the curves become closer farther into the crystalline material.

The model assumes one-dimensional conduction, a semi-infinite material, constant T_s and V , quasi-steady response, no thermal expansion, and a fixed phase-transition temperature. It does not solve the external surface energy balance, form char, or transport pyrolysis gas. The tests verify the stated reduced model, not these assumptions for a particular flight or test environment.



(a) Figure 6: analytical and finite-difference profiles.



(b) Figure 7: surface-temperature sweep.

Figure 1: Reproduced temperature profiles. Figure 6 compares the integral solution with the independent finite-difference result. Figure 7 changes T_s while holding V fixed.

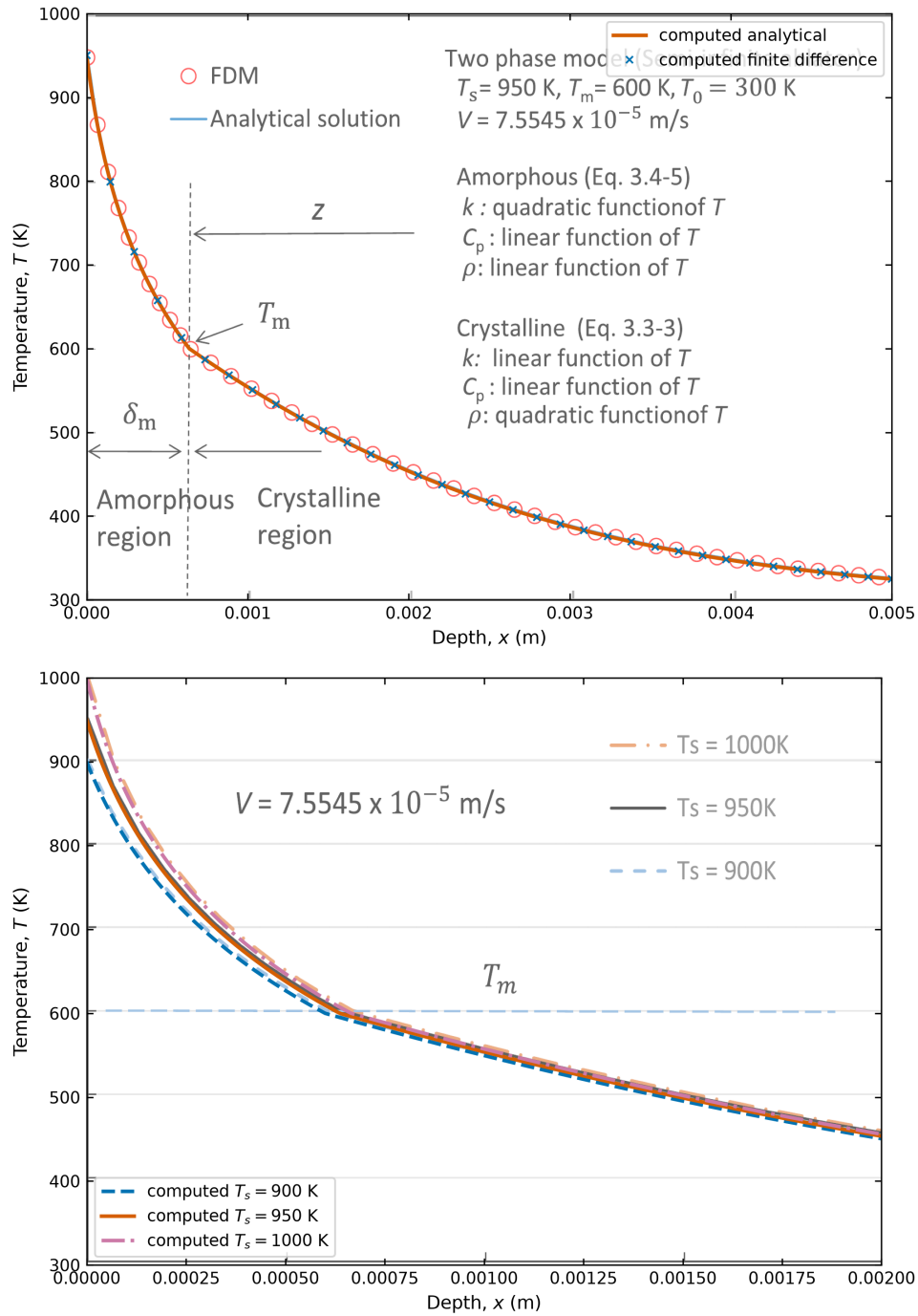


Figure 2: Calculated profiles over the rasterized published Figures 6 and 7. The overlays check registration and curve shape. Numerical errors are calculated from the independent solutions, not from image pixels.

12 Reproducibility

From the project root, run

```
uv sync
uv run pytest
uv run python symbolic_derivation.py
uv run python compare_solutions.py
uv run python -m plots.plot_figure6
uv run python -m plots.plot_figure7
uv run python -m plots.plot_paper_overlays
cd writeup
latexmk -pdf -interaction=nonstopmode -halt-on-error derivation.tex
```

References

- [1] Sumio Kato and Shoichi Matsuda. Analytical solution for the problem of one-dimensional quasi-steady non-charring ablation in a semi-infinite solid with temperature-dependent thermo-physical properties. *Thermal Science and Engineering Progress*, 31:101181, 2022.